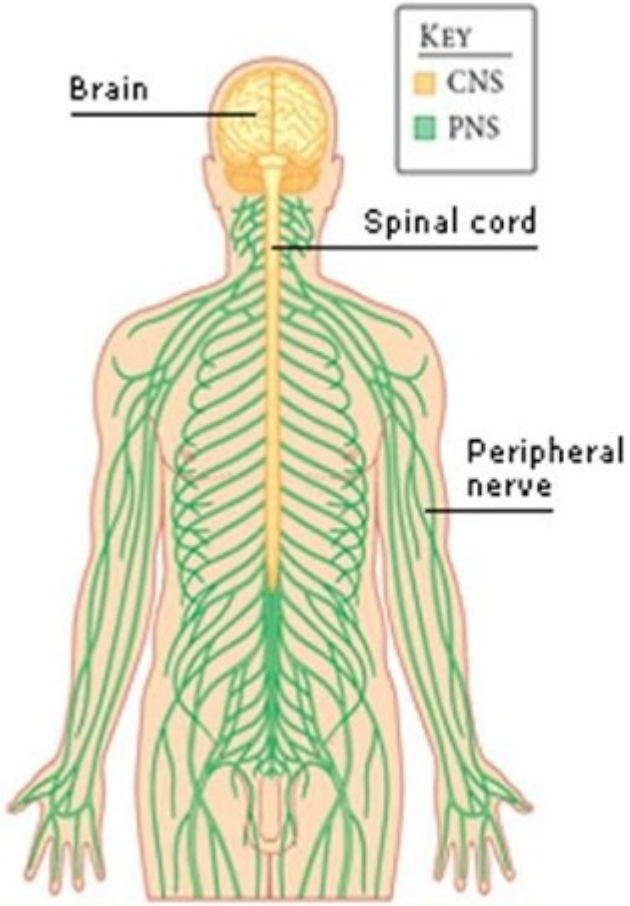
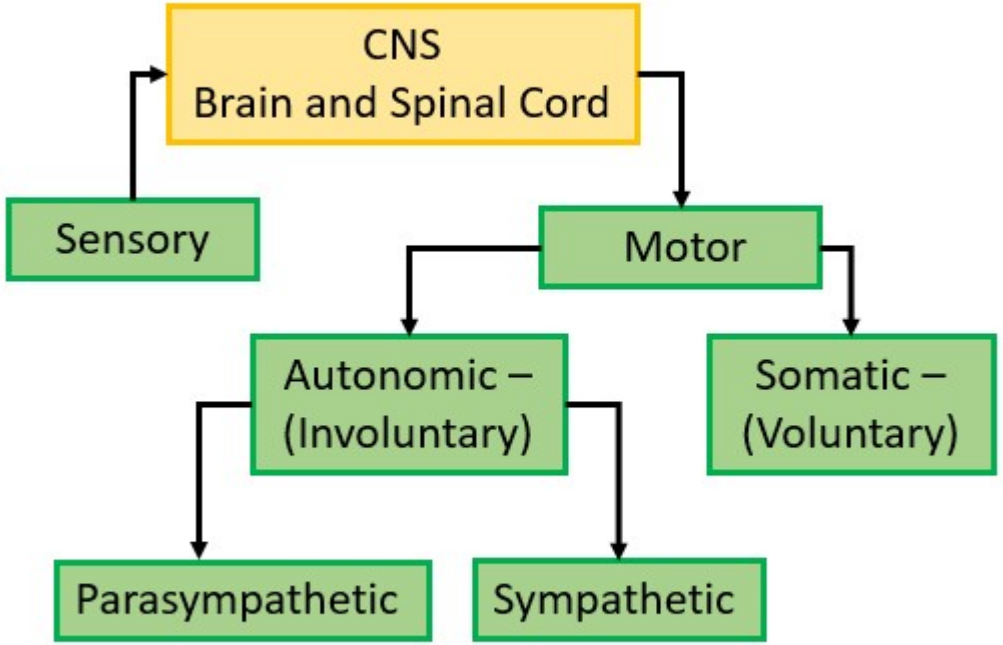
Divisions of the Nervous System - Overview
Peripheral Nervous System (PNS) - Sensory

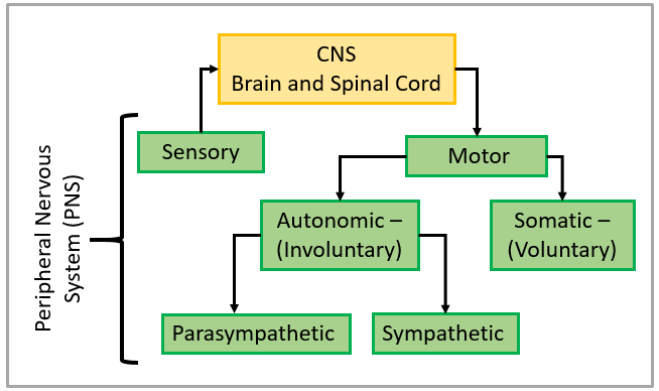




The nervous system is highly organised

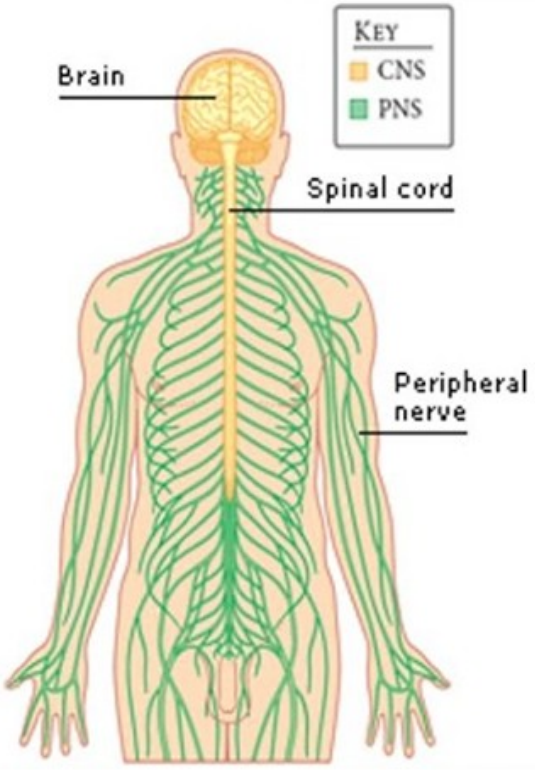
Overview

- Nervous system consists of the brain, spinal cord, and all nerves.



- Peripheral Nervous System (PNS) – we will focus on this now
  - Sensory and motor neurons, bundled into nerves
  - Take information to and from CNS
  - Clusters of nerve cell bodies (*ganglia)*, outside of CNS

- Central Nervous System (CNS) – we will focus on this after PNS
  - brain and spinal cord

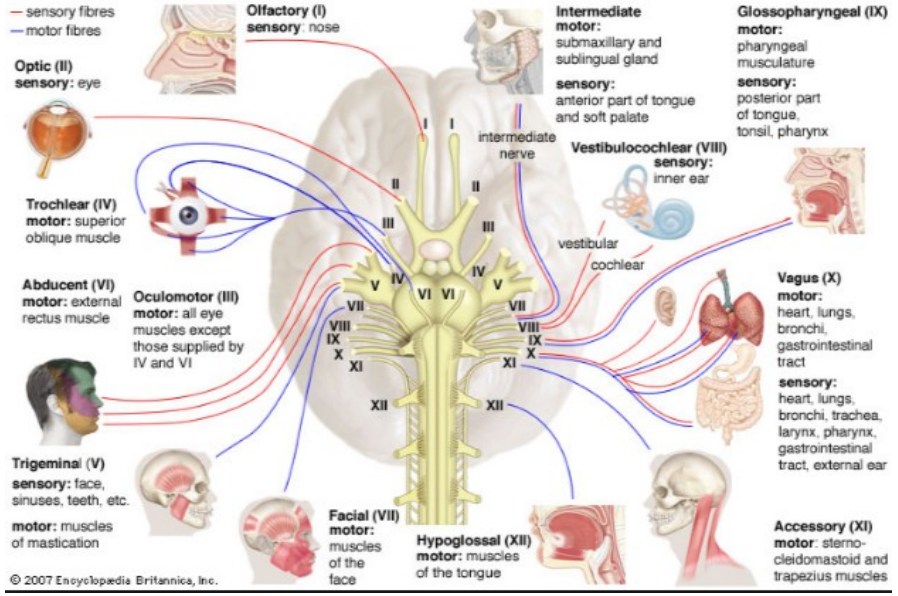


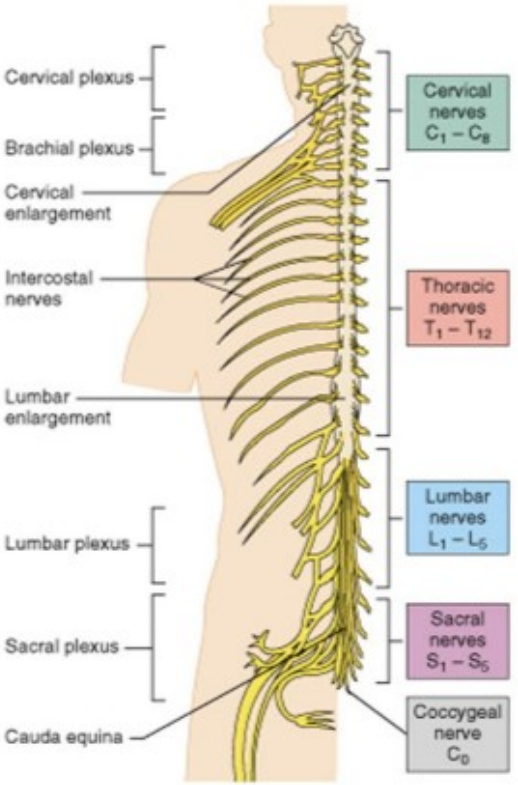
Learning Aim: List the main parts of the Nervous System

Nerves

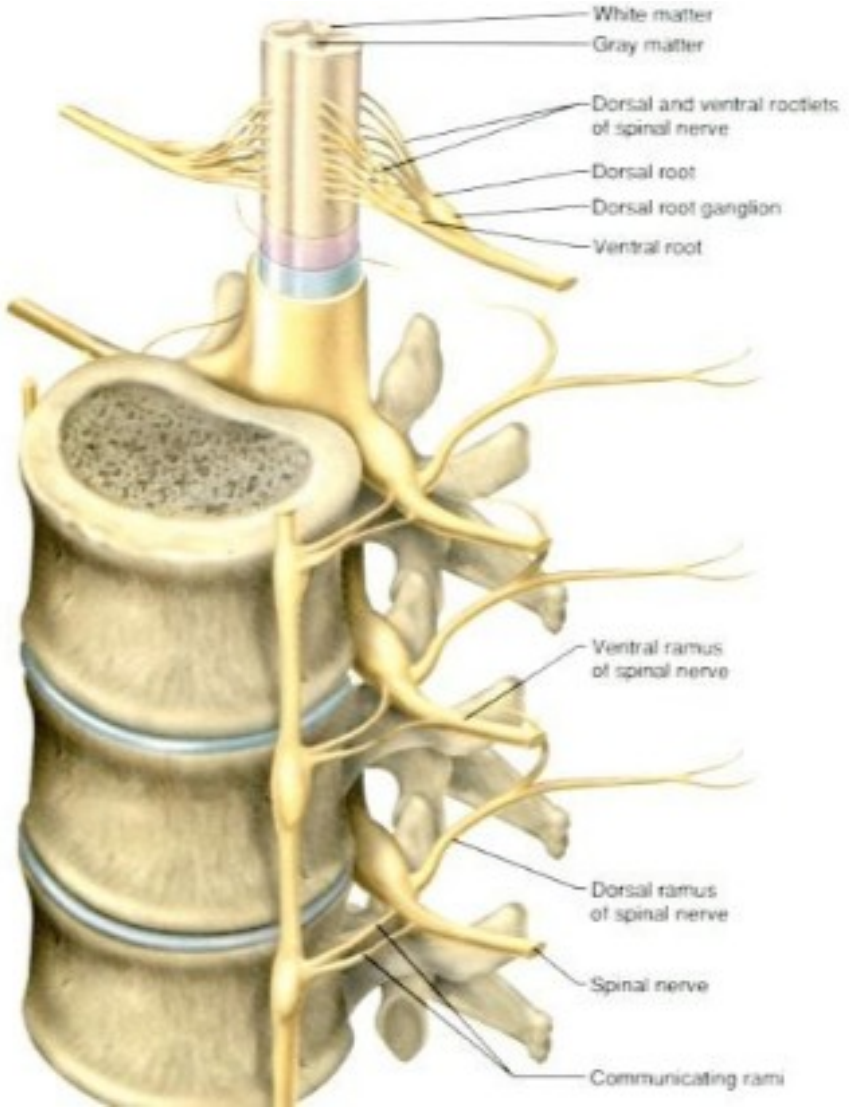
- Bundles of neurons
- Can contain both sensory and motor neurons (mixed nerves)
- Connect to brain /spinal cord
- Transmit signals to and from CNS
- Cranial Nerves (attached a base of brain):
  - 12 cranial nerves
  - Some mixed nerves, others only sensory or only motor
  - Feed directly into brain base
- Spinal Nerves (attach along spinal cord)
  - 31 pairs of spinal nerves
  - All mixed nerves
  - Attach to spinal cord by 2 roots

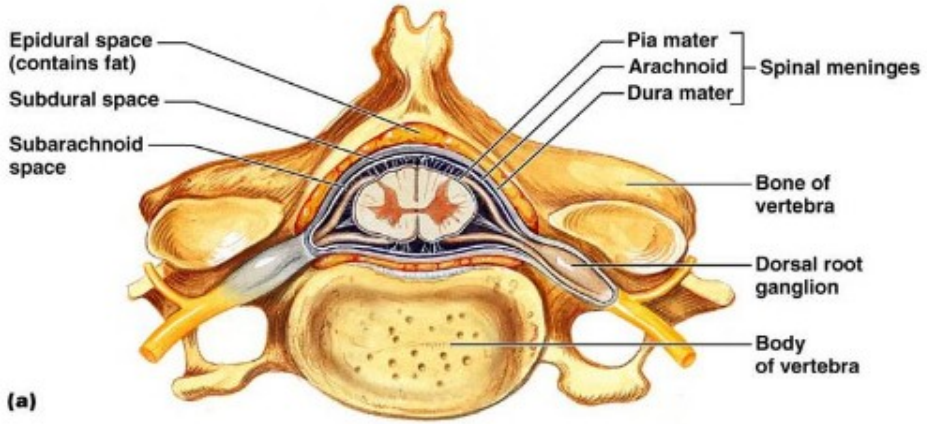
Learning Aim: Describe how PNS nerves connect with the CNS

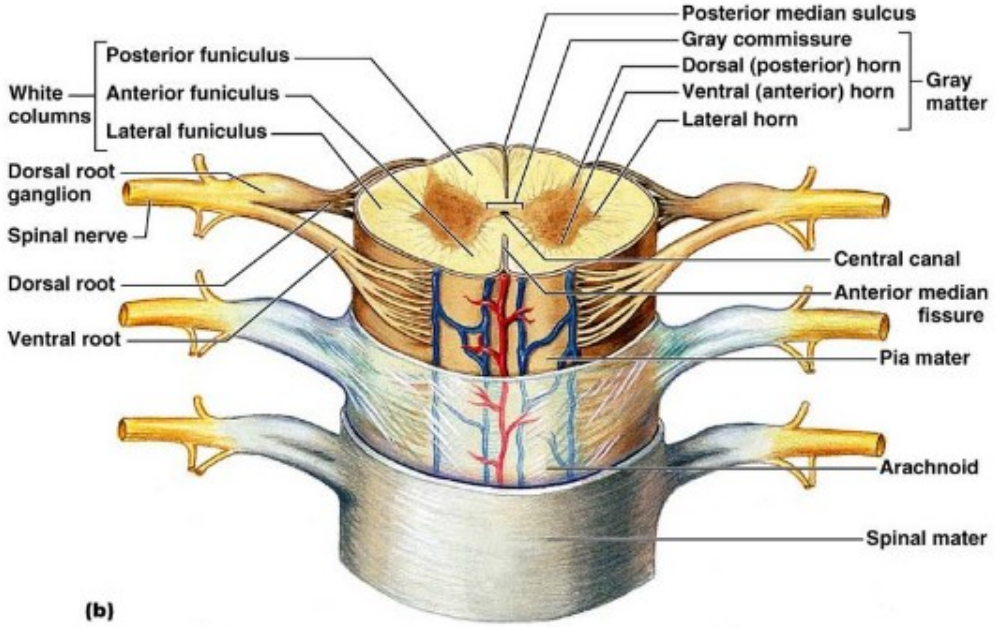




Spinal Nerve connection to Spinal Cord

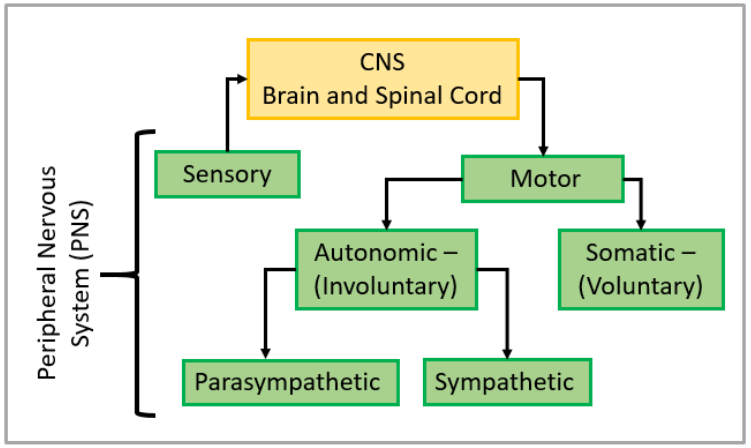


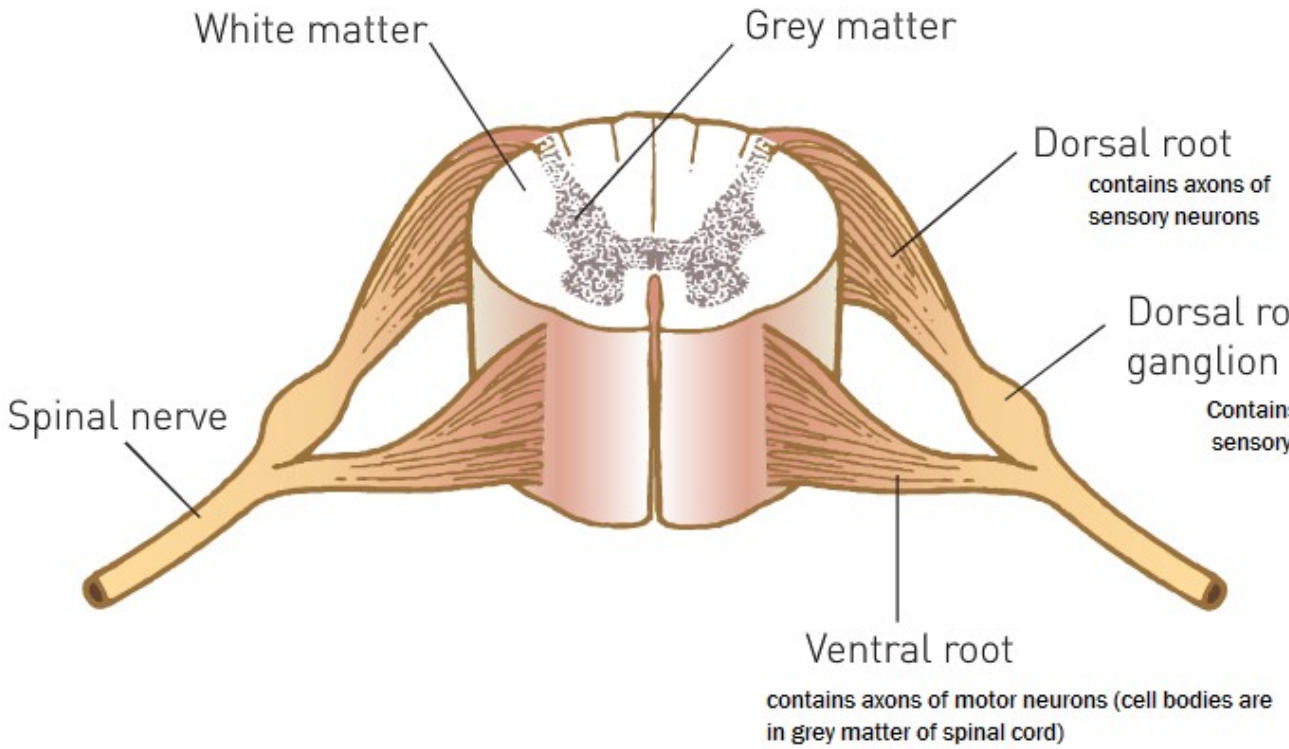


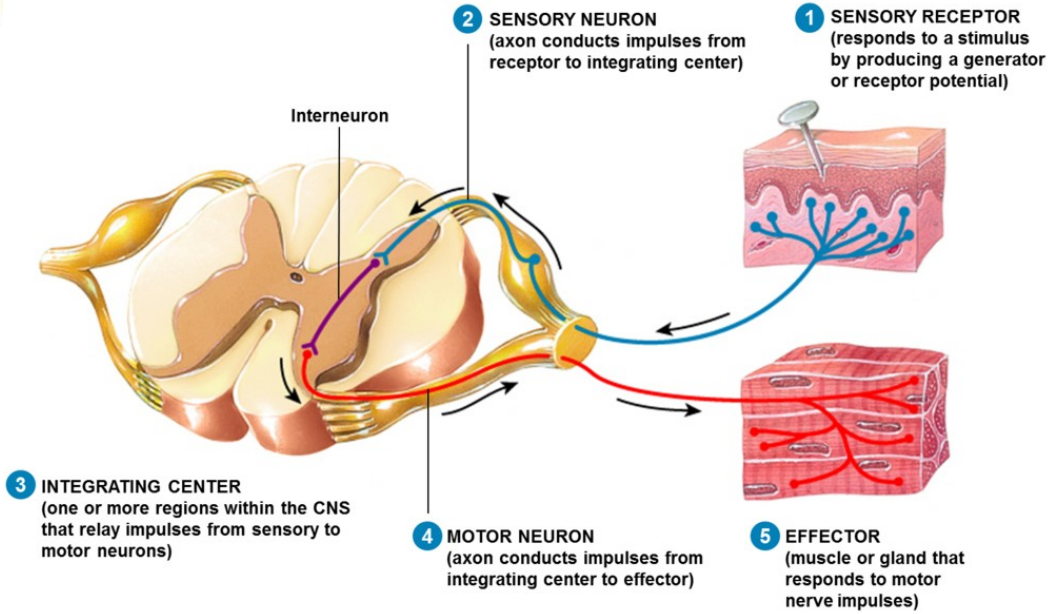


Learning Aim: Describe how PNS nerves connect with the CNS

Spinal Nerve connection to Spinal Cord

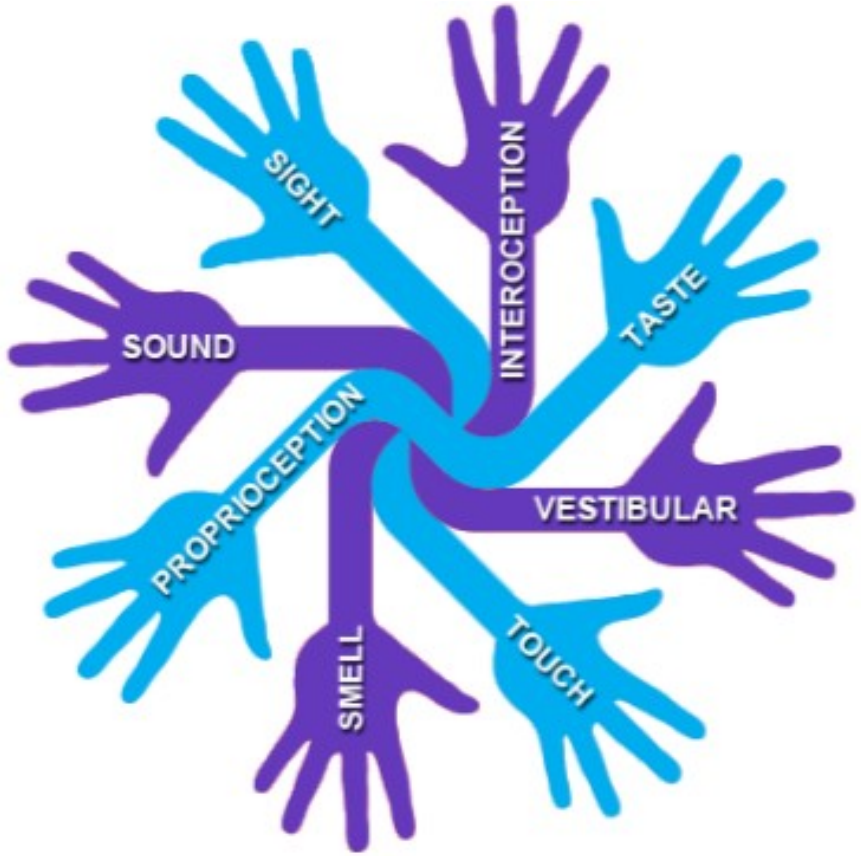


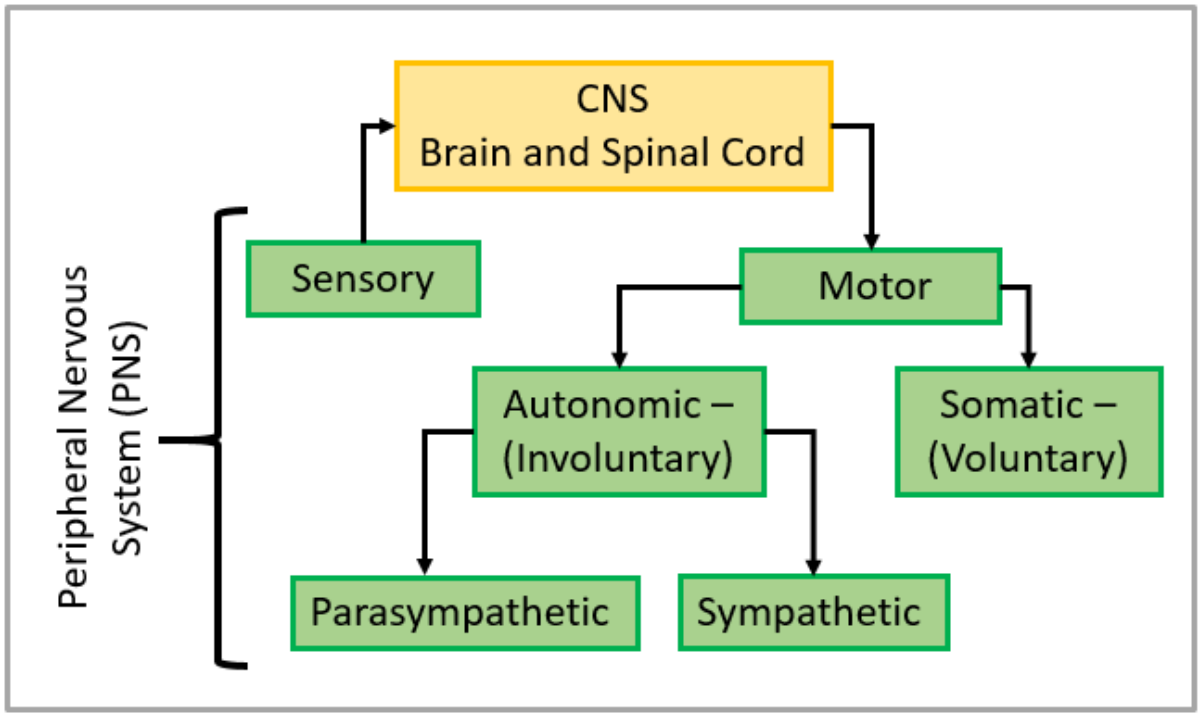




Learning Aim: Describe how PNS nerves connect with the CNS

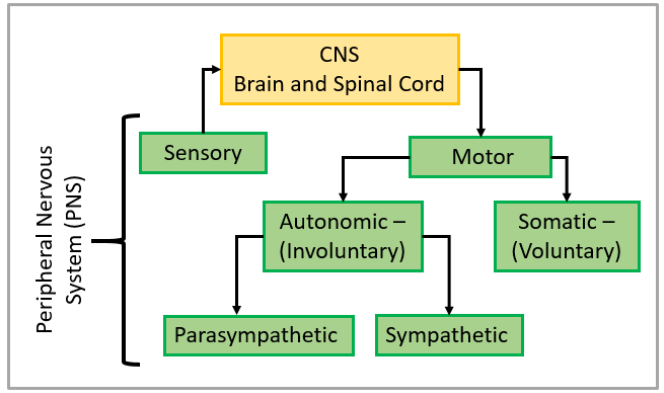
Peripheral Nervous System: Sensory (Afferent) Division





The nervous system is highly organised

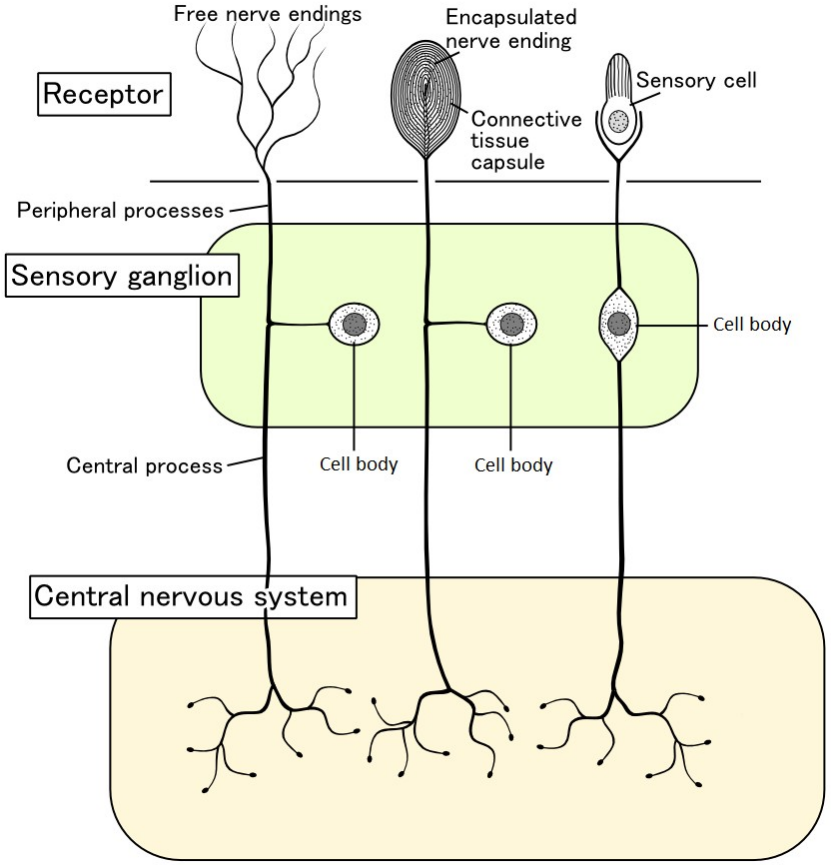
Senses and Receptors (PNS-Sensory)



1. Body receives signals at receptors/sensors.

2. Stimulated receptors send signals to sensory neurons.

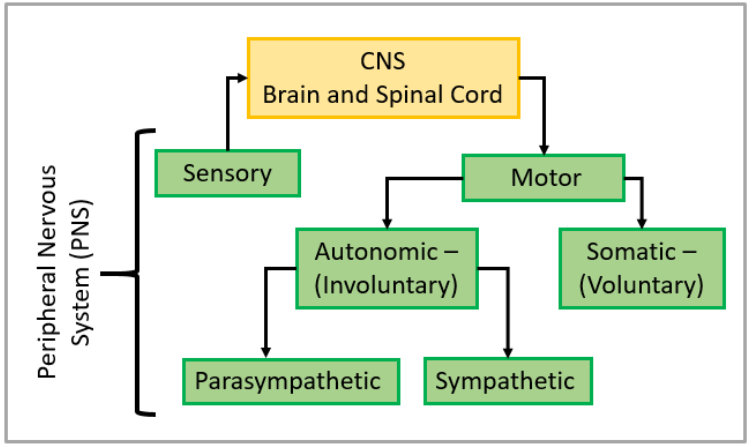
3. Signals are passed to the CNS to be integrated and acted upon.

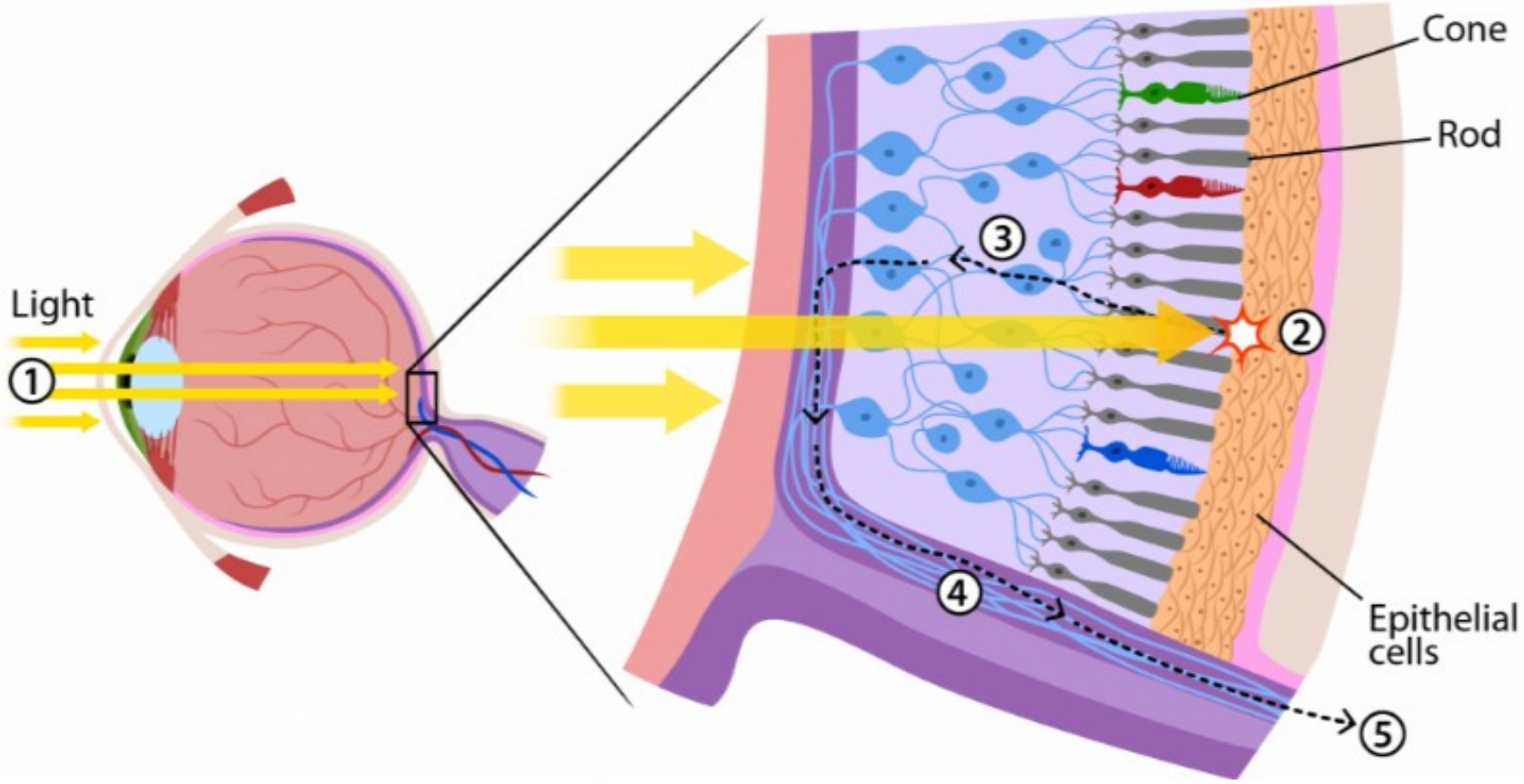


Learning Aim: Describe the function of the PNS Sensory

Sense organs (PNS-Sensory):

- Eye:
  - Rod and cone cells in retina stimulated by light
  - Transmit signals via optic nerve to brain

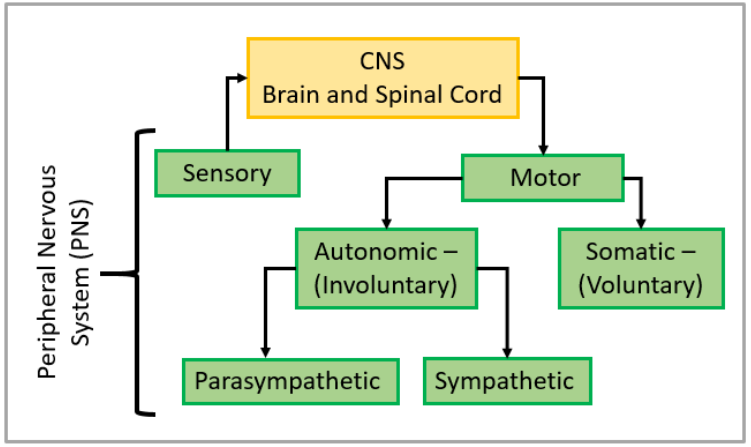


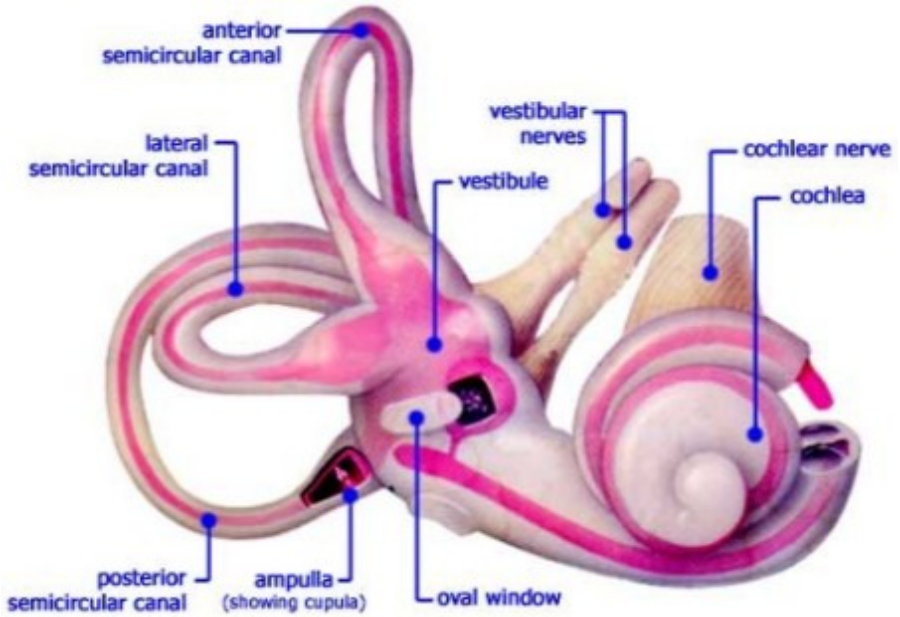


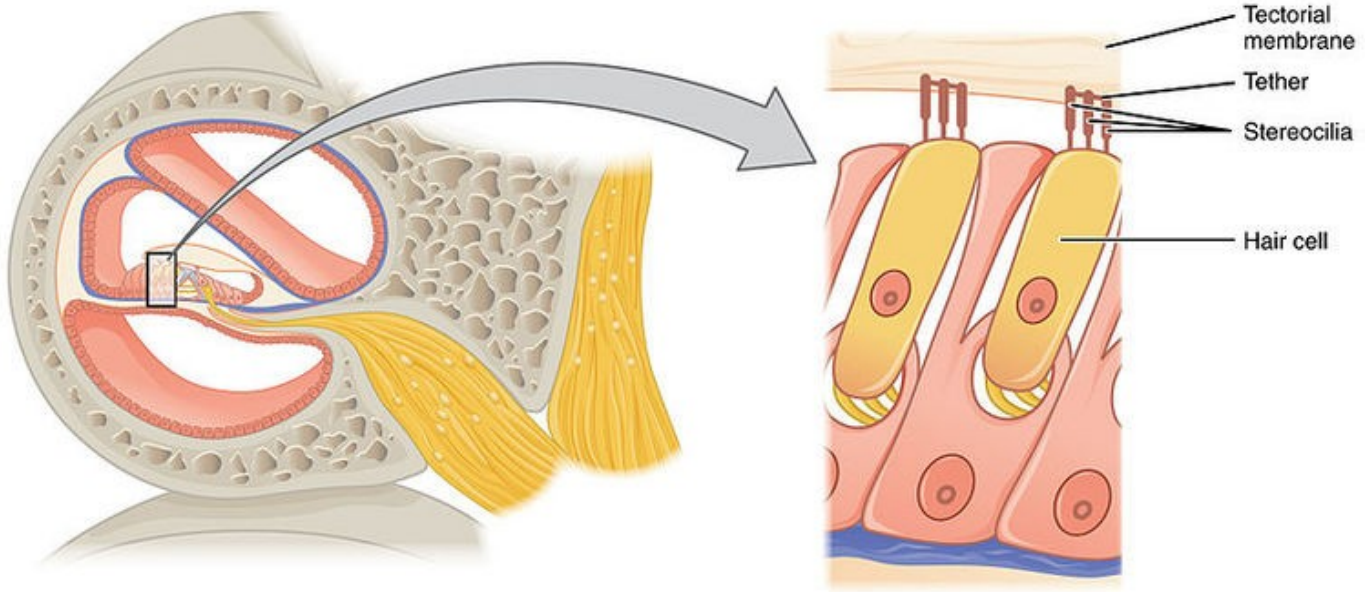
Learning Aim: Briefly describe how the sense organs receive information and pass it via the sensory PNS to the CNS

Sense Organs:

- Ear:
  - Cochlea contains fluid and fine hair-like sensors that detect sound waves.
  - Transmit signals to brain via auditory nerve
  - Vestibular system series of canals and specialised sensory cells that detect position of head – balance.
  - Signals sent to brain via vestibular nerve





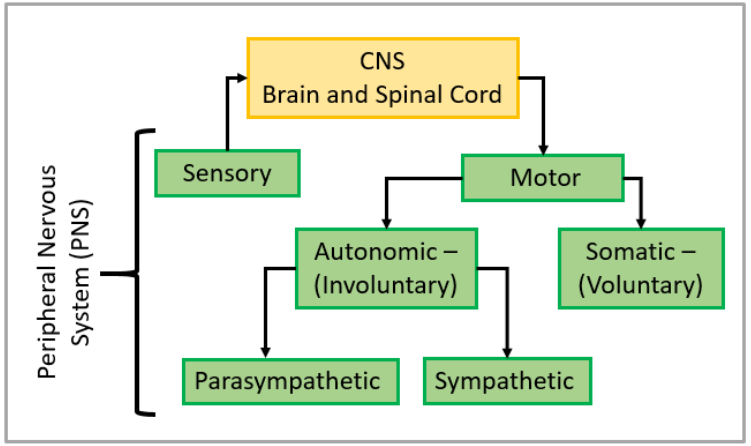


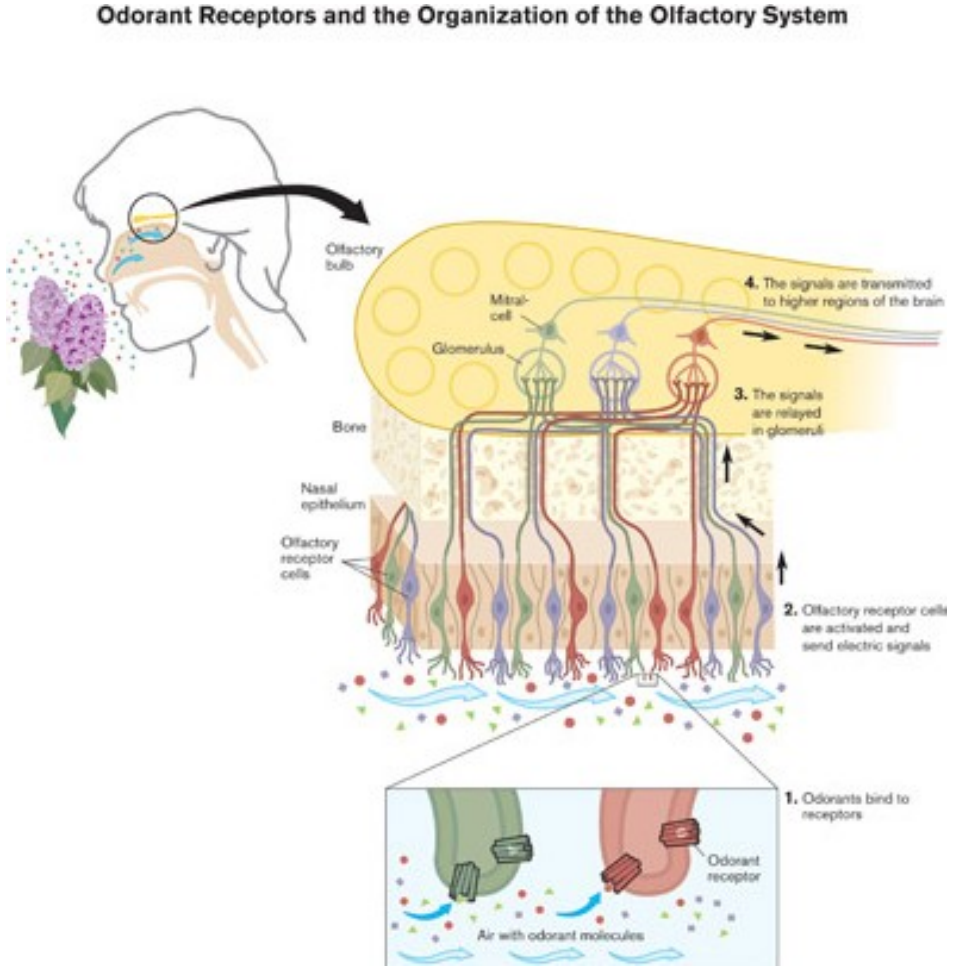
Learning Aim: Briefly describe how the sense organs receive information and pass it via the sensory PNS to the CNS

Sense Organs:

Nose (olfactory)

- Specialised cells in nasal cavity send signals to brain via olfactory bulb and olfactory nerve.



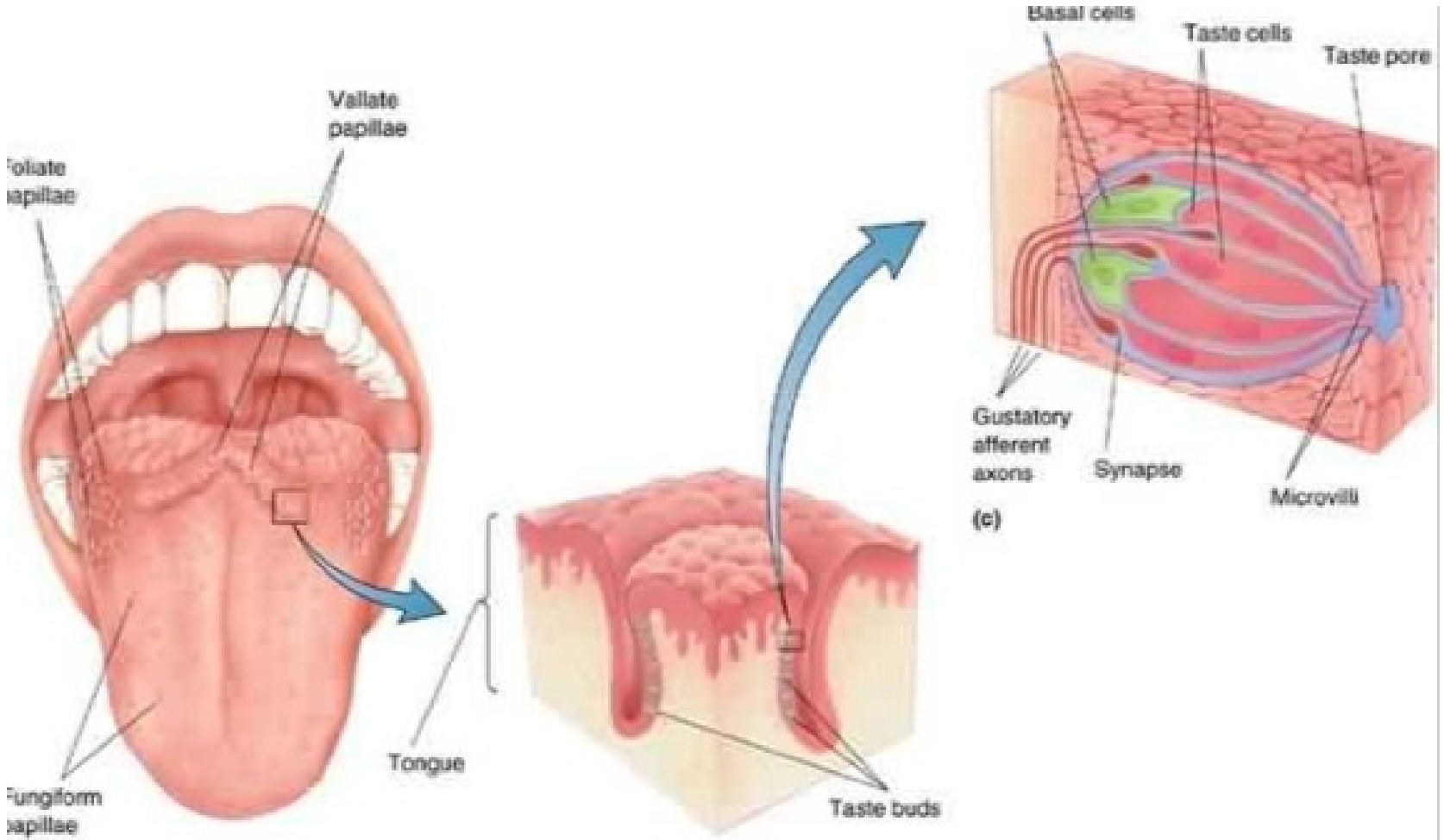


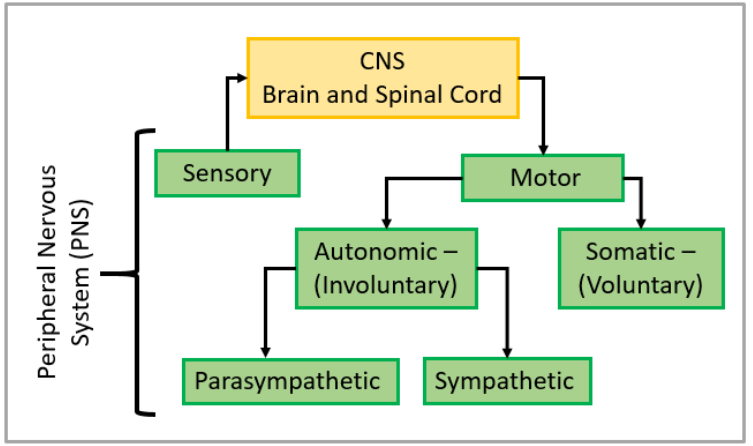
Learning Aim: Briefly describe how the sense organs receive information and pass it via the sensory PNS to the CNS

Sense Organs:

Taste

- Specialised cells within epithelium of tongue send signals via facial nerve and other fibres

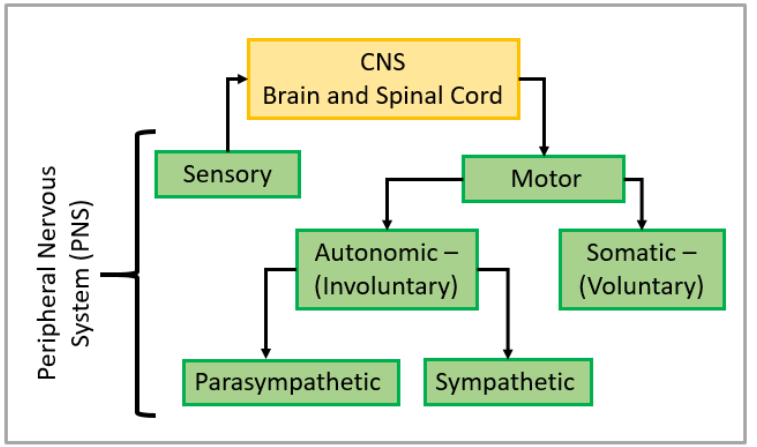


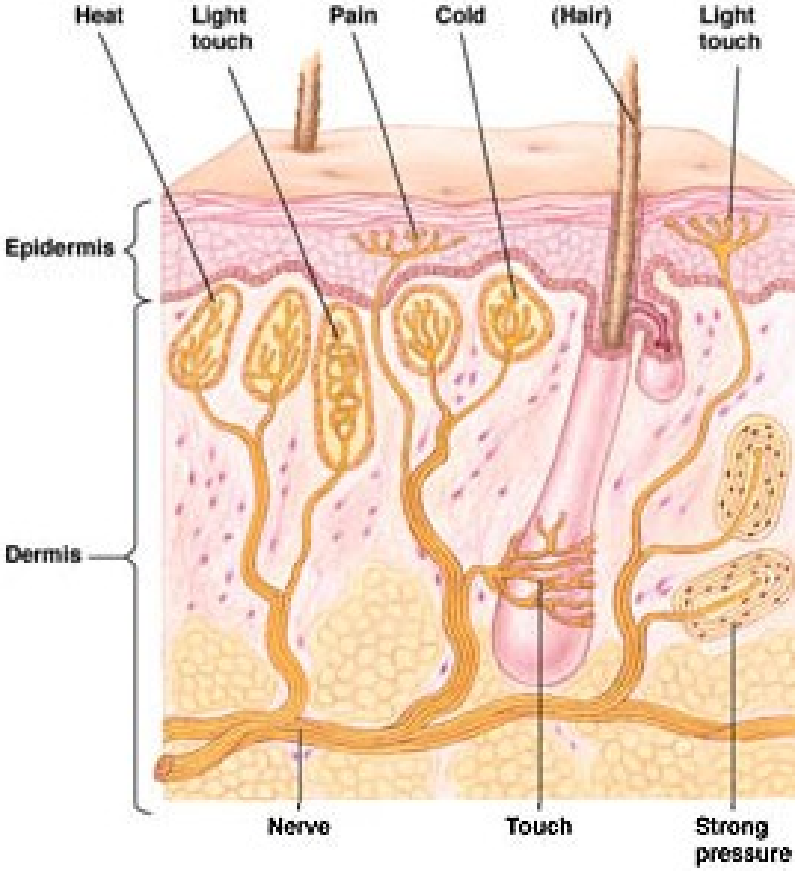


Learning Aim: Briefly describe how the sense organs receive information and pass it via the sensory PNS to the CNS

Senses and receptors

- Touch
  - Specialised cells in the skin
    - Heat (thermoreceptors)
    - light pressure
    - pain
    - cold
    - strong pressure
  - Send signals via sensory nerves to the CNS

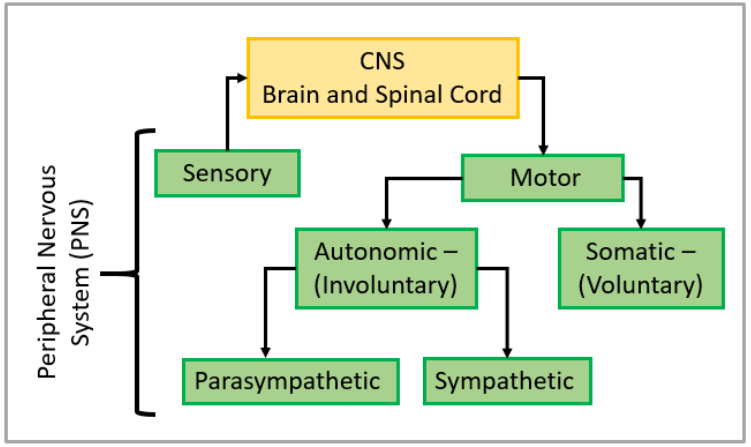


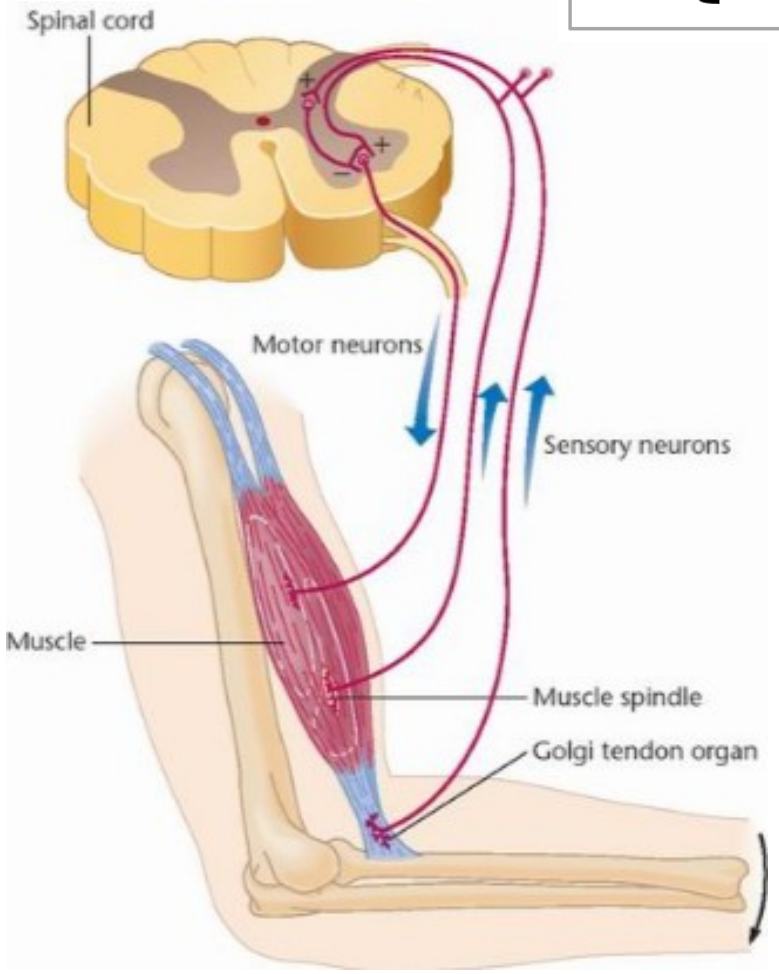


Learning Aim: Describe the receptors involved in touch

Senses and Receptors

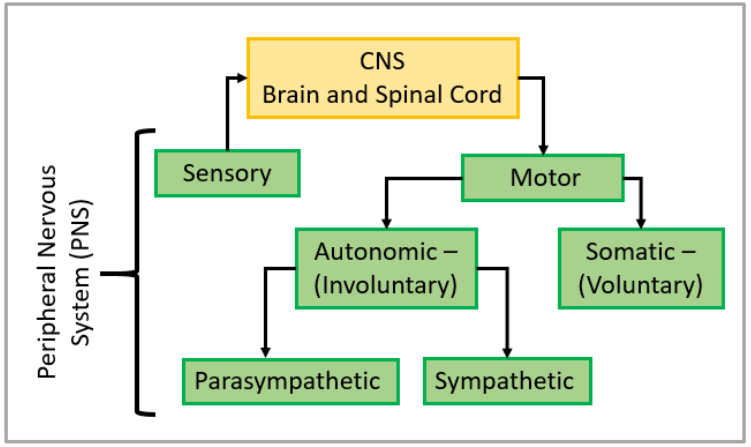
- Proprioception
  - Receptors located in muscles and joints provide information about body positioning



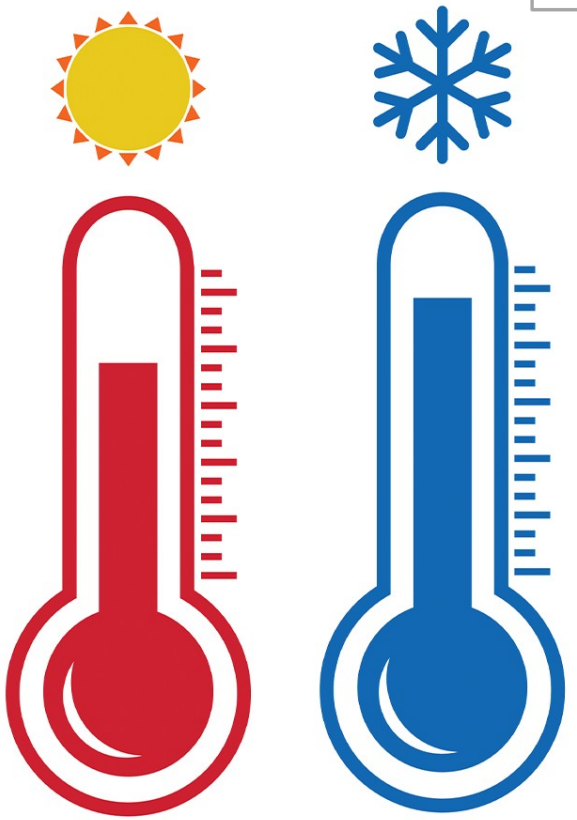


Learning Aim: Describe the receptors involved in proprioception

Types of receptors:



- Thermoreceptors
  - Detect heat and cold
  - Found in skin (detect surface temp)
  - Found in hypothalamus (detect temp of blood – core temp)
  - Information sent to CNS:
    - Hypothalamus regulates temperature
    - Brain: conscious perception of temperature



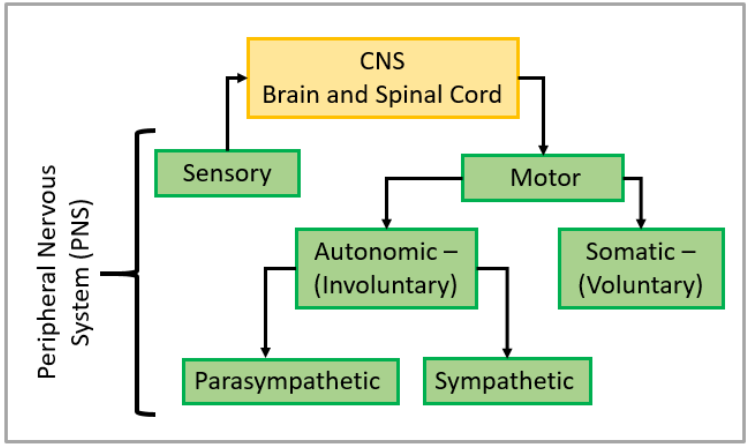
Learning Aim: List different sensors in the body, where they are located and what they sense.

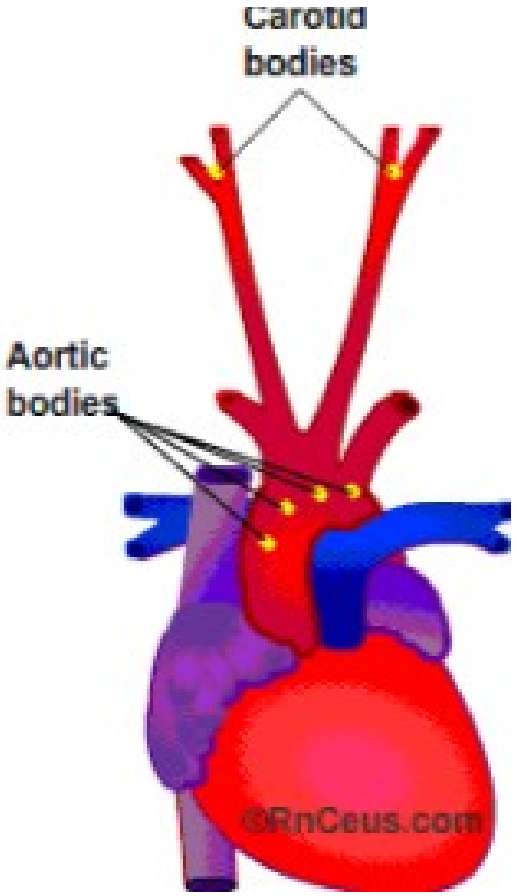
Types of receptors:

- Osmoreceptors
  - In hypothalamus
  - Detect osmotic pressure (related to ion concentration), monitoring water homeostasis

- Chemoreceptors
  - Different types of chemoreceptors
  - Stimulated by particular chemicals
  - In nose and mouth – smell and taste
  - In blood vessels (aorta and carotids): detect blood pH, O_2 and CO_2 concentration.

- Baroreceptors
  - Detect Blood Pressure
  - In carotids and aortic arch

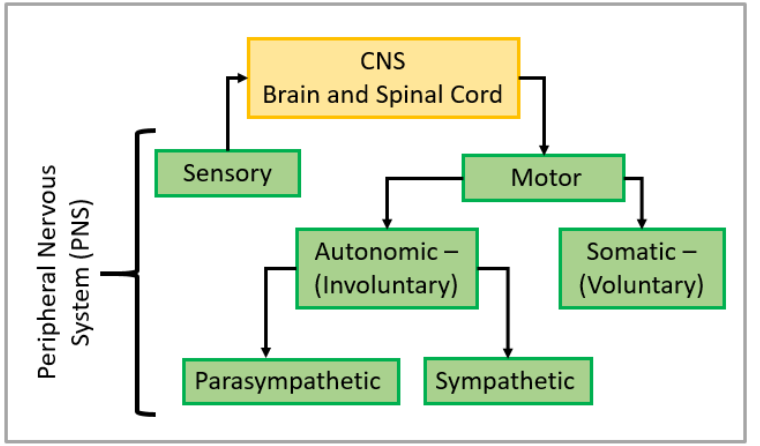




Learning Aim: List different sensors in the body, where they are located and what they sense.

Types of receptors:

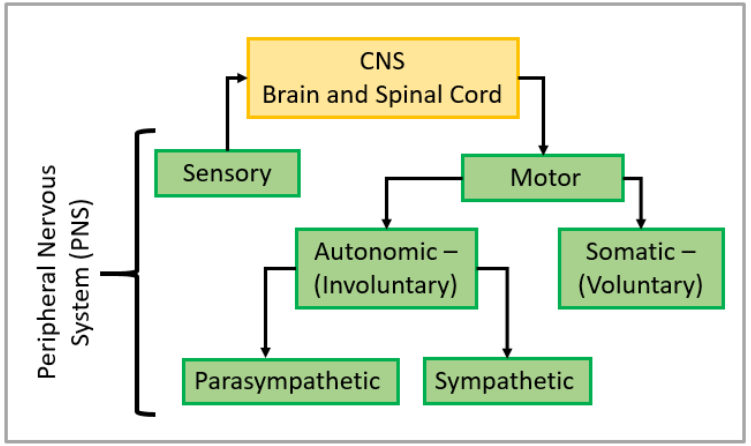
- Pain receptors (nociceptors)
  - In almost all organs
    - Many in skin and mucous membranes
    - None in brain
  - Stimulated by tissue damage, or risk of tissue damage, and carry signals to brain.

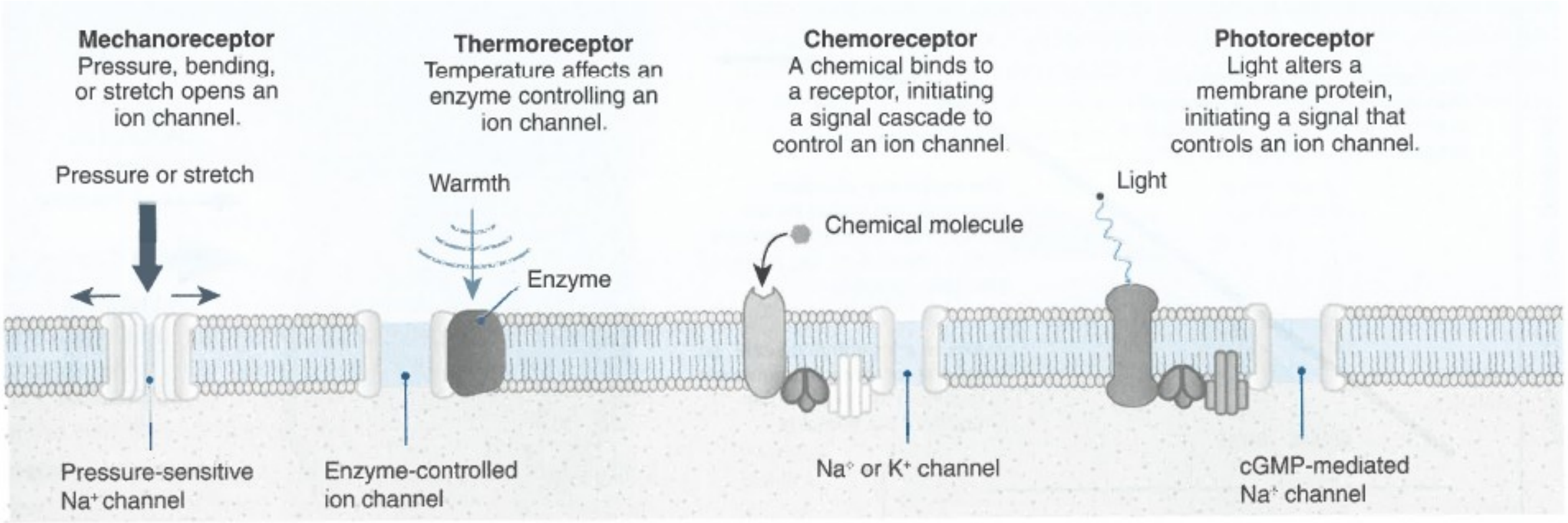




Learning Aim: List different sensors in the body, where they are located and what they sense.

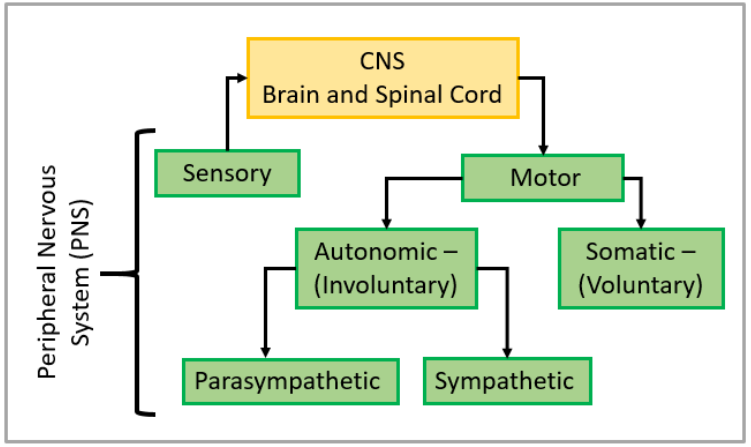
Some examples of how receptors work at a membrane level…

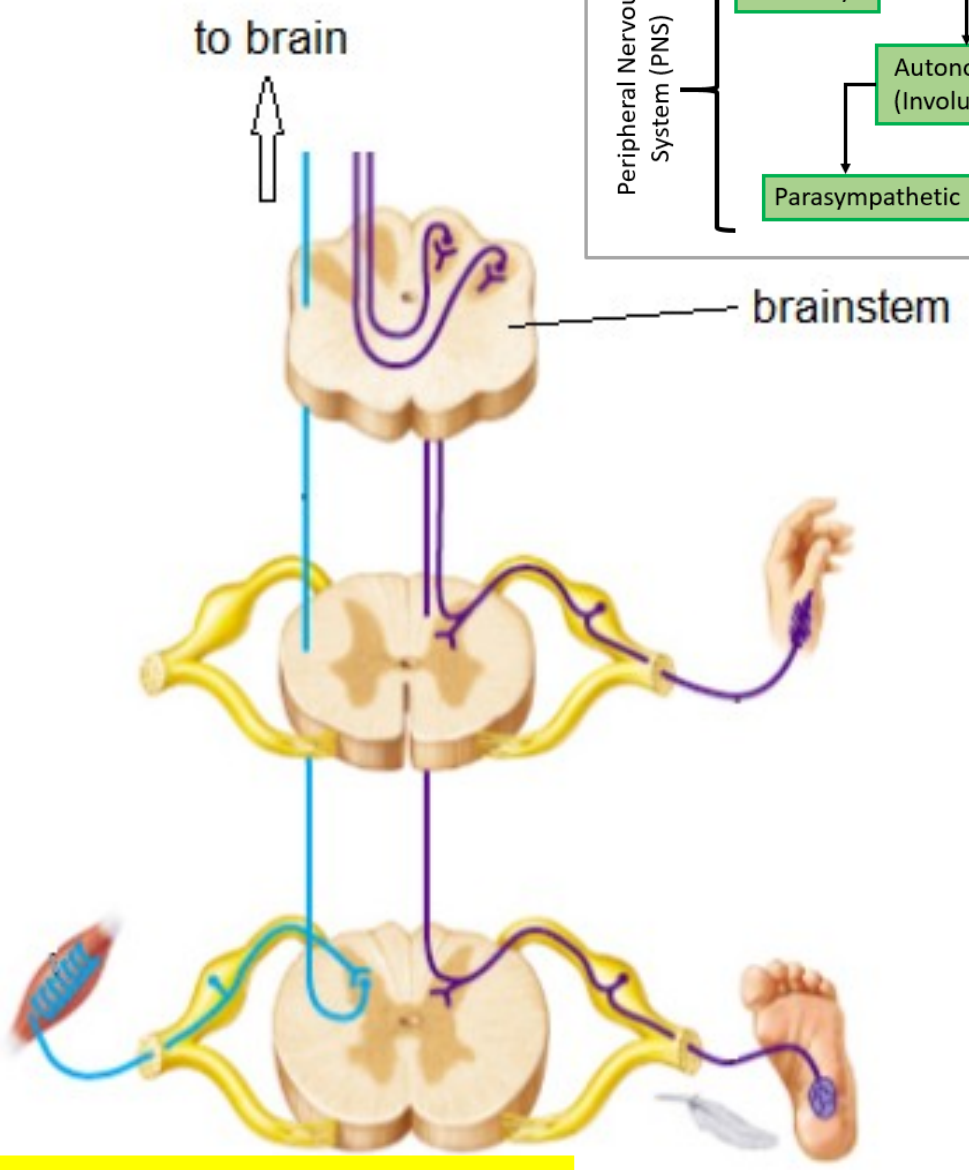




Sensory neurons

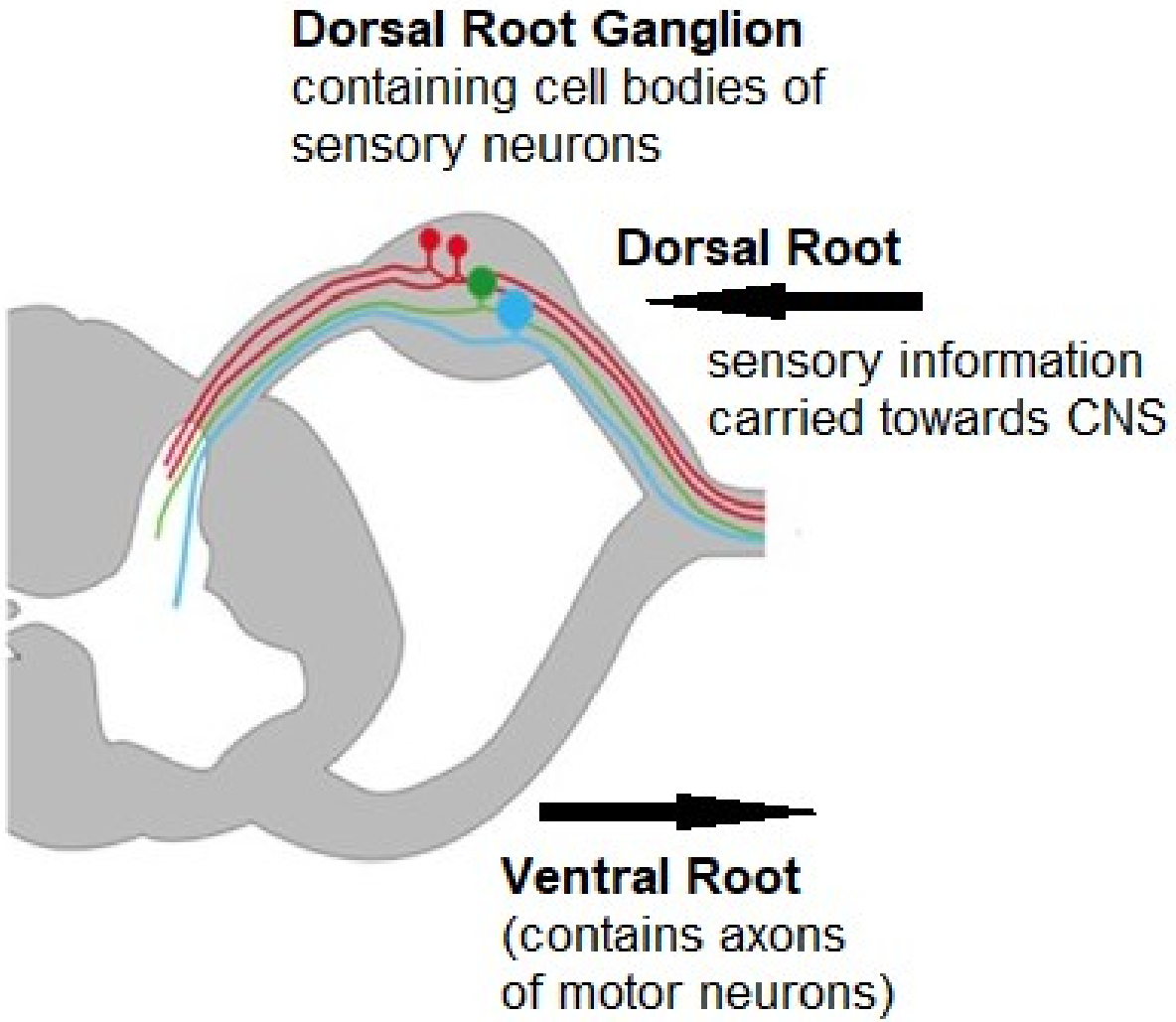
- Stimulated by sense organs/receptors
- Bundled into nerves
  - Some nerves contain only sensory fibres
  - Other (mixed) nerves contain some sensory fibres, but also motor fibres
- Carry signals to spinal cord (spinal nerves) or base of brain (cranial nerves)
  - Sensory neurons enter via dorsal root of spinal cord
  - Cell bodies of sensory neurons are located in dorsal root ganglion
- Spinal cord relays information to brain
- Information provided about body's current state and functions

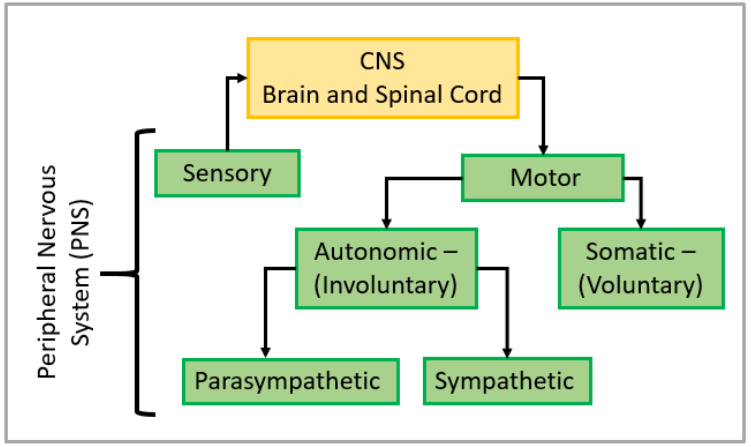




Learning Aim: Describe the organisation of sensory neurons and their pathway into the spinal cord.

Sensory neurons





Learning Aim: Describe the organisation of sensory neurons and their pathway into the spinal cord.